is the formation of abnormal chemical bonds between adjacent pyrimidine molecules in the same strand, or between pyrimidines on the opposite strands, of the double helix. This bonding is induced mostly between adjacent thymines, forming what are called thymine dimmers (Figure 12.10), usually designated TT. This unusual pairing produces a bulge in the DNA strand and disrupts the normal pairing of T's (thymines) with corresponding A's(adenines) on the opposite strand. If UV induced genetic damage is not repaired, mutations or cell death may result.

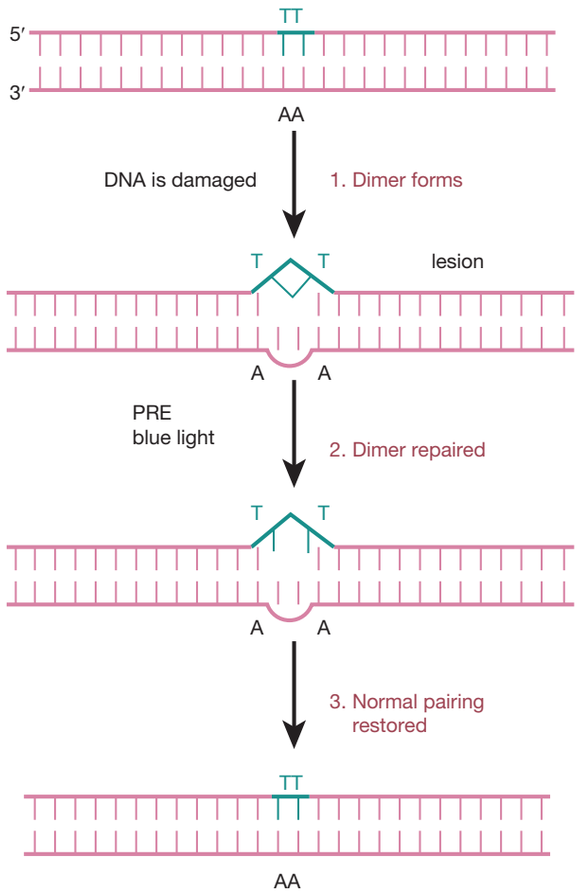


Figure 12.10: UV induced DNA damage

Chemical Mutagens

Chemical mutagens include both naturally occurring chemicals and synthetic substances. These mutagens can be grouped into different classes on the basis of their mechanism of action. They are

i. **Base analogs** are bases that are similar to the bases normally found in DNA. Like normal bases, base analogs exist in normal and rare tautomeric states. In each of the two states, the base analog pairs with a different normal base in DNA. Because base analogs are similar to the normal nitrogen bases, they may be incorporated into DNA in place of normal bases. E.g. 5 – bromouracil (5-BU). If 5-BU is incorporated in its more common normal state, it pairs with adenine. If it changes into its rare state during replication, it pairs with guanine instead. In the next round of replication, the 5-BU – G base pair is resolved into a C – G base pair instead of the T – A base pair. By this process a transition mutation is produced, from TA to CG (Figure 12.11).



Not all base analogs are mutagens. For example, AZT (Azidothymidine), one of the approved drugs given to patients with AIDS, is an analog of thymidine, but it is not a mutagen, because it does not cause base pair changes.

ii. **Base Modifying Agents** are chemical that act as mutagens by modifying the chemical structure and properties of bases. The three types of mutagens that work in this way are

1. A deaminating agent e.g.: Nitrous acid removes amino groups (- NH_2) from the bases guanine, cytosine, and

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